

Signal Processing

Cheat sheet

Elementary Signals

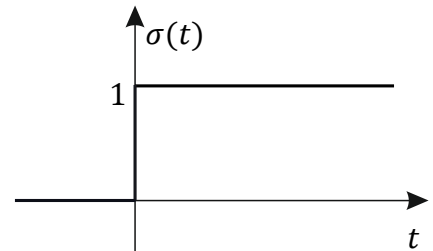
Dirac Delta Distribution

$$\delta(t) = 0, \text{ for } t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \text{Area} = 1$$

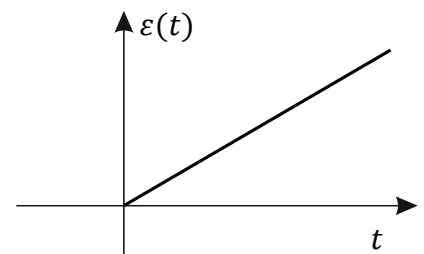
Heaviside Step Function

$$\sigma(t) = \begin{cases} 1, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$



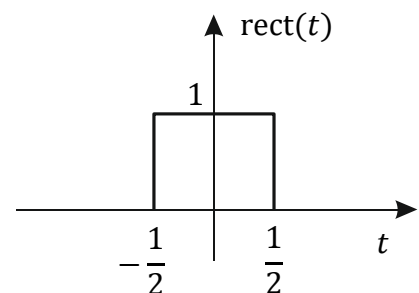
Ramp Function

$$\varepsilon(t) = t \cdot \sigma(t)$$



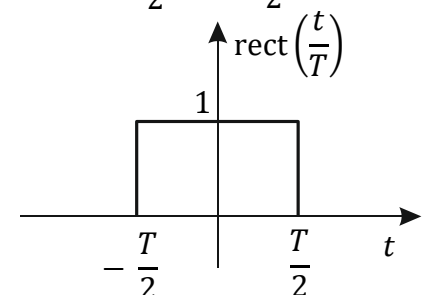
Rectangular Function

$$\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$



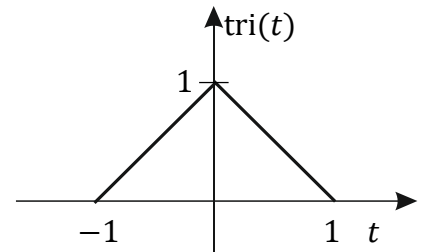
time-scaled

$$\text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & |t| < \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$



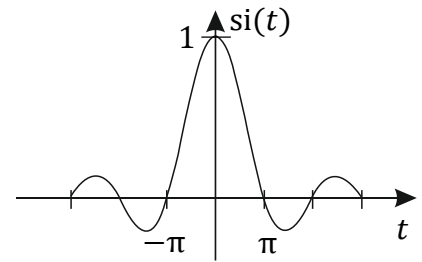
Triangular Function

$$\text{tri}(t) = \begin{cases} 1 - |t|, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$



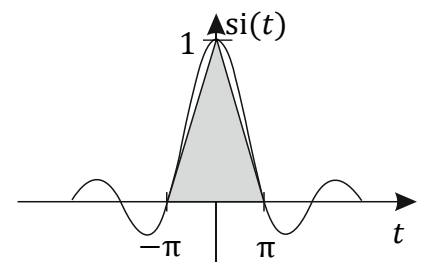
Sinc Function

$$\text{si}(t) = \begin{cases} \frac{\sin(t)}{t}, & \text{for } t \neq 0 \\ 1, & \text{for } t = 0 \end{cases}$$



$$\int_{-\infty}^{\infty} \text{si}(t) dt = \pi$$

Note: Area of the triangle
is equal to the integral
of the sinc function.



Fourier Series

Sine-Cosine Form

$$y(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cdot \cos(\omega_0 nt) + b_n \cdot \sin(\omega_0 nt)] \quad \text{with} \quad \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

Exponential Form

$$y(t) = \sum_{-\infty}^{\infty} c_n \cdot e^{j\omega_0 nt}$$

Calculation of the coefficients for the sine-cosine form and the exponential form

Coefficient for the DC Component	Sine-Cosine Form	Exponential Form
$a_0 = \frac{2}{T} \int_T y(t) dt$	$a_n = \frac{2}{T} \int_T y(t) \cdot \cos(\omega_0 nt) dt$	$c_n = \frac{1}{T} \int_T y(t) \cdot e^{-j\omega_0 nt} dt$
	$b_n = \frac{2}{T} \int_T y(t) \cdot \sin(\omega_0 nt) dt$	

Calculation of the sine-cosine form based on the complex form

$$c_0 = \frac{a_0}{2}; \quad c_n = \frac{1}{2}(a_n - jb_n); \quad c_{-n} = c_n^* = \frac{1}{2}(a_n + jb_n)$$

c_0 : DC Component

Calculation of the complex form based on the sine-cosine form

$$a_0 = 2c_0; \quad a_n = c_n + c_{-n} = c_n + c_n^*; \quad b_n = j(c_n - c_{-n}) = j(c_n - c_n^*)$$

Euler

$$\sin(x) = \frac{1}{2j} \cdot (e^{jx} - e^{-jx})$$

$$\cos(x) = \frac{1}{2} \cdot (e^{jx} + e^{-jx})$$

$$e^{jx} = \cos(x) + j \cdot \sin(x)$$

Fourier Transform

$X(j\omega) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-j\omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) \cdot e^{j\omega t} d\omega$
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Property, Theorem	Time Domain	Frequency Domain
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Differentiation	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
Time Domain Scaling	$x(at), a \neq 0$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
Frequency Domain Scaling	$\frac{1}{ b } x\left(\frac{t}{b}\right)$	$X(bj\omega), b \neq 0$
Convolution	$x(t) * y(t)$	$X(j\omega) \cdot Y(j\omega)$
Multiplication	$x(t) \cdot y(t)$	$\frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Time Domain Shift	$x(t + a)$	$e^{j\omega a} X(j\omega)$
Frequency Domain Shift	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$

Time Domain	Frequency Domain
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$\dot{\delta}(t)$	$j\omega$
$\sigma(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\text{rect}(at)$	$\frac{1}{ a } \text{si}\left(\frac{\omega}{2a}\right)$
$\text{tri}(at)$	$\frac{1}{ a } \cdot \text{si}^2\left(\frac{\omega}{2a}\right)$
$\text{si}(at)$	$\frac{\pi}{ a } \text{rect}\left(\frac{\omega}{2a}\right)$
$\frac{1}{t}$	$-j\pi \cdot \text{sgn}(\omega)$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$\sin(\omega_0 t)$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$

z Transform

$$\text{Definition: } X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Sequence $x[n]$	z-Domain $X(z)$	ROC
$\delta[n]$	1	$z \in \mathbb{C}$
$\delta[n - n_0]$	z^{-n_0}	$0 < z < \infty$
$\sigma[n]$	$\frac{z}{z - 1}$	$ z > 1$
$n \cdot \sigma[n]$	$\frac{z}{(z - 1)^2}$	$ z > 1$
$\alpha^n \cdot \sigma[n]$	$\frac{z}{z - \alpha}$	$ z > \alpha $
$\binom{n}{m} \alpha^{n-m} \cdot \sigma[n]$	$\frac{z}{(z - \alpha)^{m+1}}$	$ z > \alpha $
$\alpha^n \cdot \sigma[-n - 1]$	$\frac{-z}{z - \alpha}$	$ z < \alpha $
$\alpha^{ n }, \quad \alpha < 1$	$\frac{z \left(\alpha - \frac{1}{\alpha} \right)}{(z - \alpha) \left(z - \frac{1}{\alpha} \right)}$	$ \alpha < z < \left \frac{1}{\alpha} \right $
$\sin(\Omega_0 n) \cdot \sigma[n]$	$\frac{z \cdot \sin(\Omega_0)}{z^2 - 2z \cos(\Omega_0) + 1}$	$ z > 1$
$\cos(\Omega_0 k) \cdot \sigma[n]$	$\frac{z [z - \cos(\Omega_0)]}{z^2 - 2z \cos(\Omega_0) + 1}$	$ z > 1$
$\frac{1}{n!} \cdot \sigma[n]$	$e^{\frac{1}{z}}$	$ z > 0$

Property, Theorem	$x[n]$	$X(z)$
Linearity	$\alpha \cdot x_1[n] + \beta \cdot x_2[n]$	$\alpha \cdot X_1(z) + \beta \cdot X_2(z)$
Time Shifting	$x[n - i]$ for $i > 0$	$z^{-i} \cdot X(z)$
Time Shifting	$x[n + i]$ for $i > 0$	$z^i \cdot X(z) - z^i \cdot \sum_{m=0}^{i-1} x[m] \cdot z^{-m}$
Scaling in z-Domain / Exponentiation	$a^n \cdot x[n]$	$X\left(\frac{z}{a}\right)$
Derivative in the z-Domain	$n \cdot x[n]$	$-z \cdot \frac{d}{dz} X(z)$
First Difference	$x[n] - x[n - 1]$	$\frac{z - 1}{z} \cdot X(z)$
Summation	$\sum_{n=0}^N x[n]$	$\frac{z}{z - 1} \cdot X(z)$
Convolution	$x_1[n] * x_2[n]$	$X_1(z) \cdot X_2(z)$

Discrete Time Systems

Sequence:

$$x[k] = x(t) \sum_{-\infty}^{\infty} \delta(t - k \cdot T_S)$$

„Standardized“ Difference Equation:

$$y[n] = \sum_{i=0}^N b_i x[n - i] - \sum_{i=1}^M a_i y[n - i]$$

Laplace Transform

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt, s = \sigma + j\omega, f(t): \text{Time Fct.}, F(s): \text{Laplace Fct.}$$

Property, Theorem	Time Domain	Laplace Domain
Linearity	$c_1 \cdot f_1(t) + c_2 \cdot f_2(t)$	$c_1 \cdot F_1(s) + c_2 \cdot F_2(s)$
Time Scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time Shifting	$f(t-a) \cdot \sigma(t-a)$	$e^{-as} \cdot F(s)$
Time Shifting	$f(t+a) \cdot \sigma(t)$	$e^{as} \left(F(s) - \int_0^a f(t)e^{-st} dt \right)$
Frequency Shifting	$e^{-at} \cdot f(t)$	$F(s+a)$
Differentiation Time Domain	$f^n(t)$	$s^n \cdot F(s) - s^{n-1} \cdot f(0) - s^{n-2} \cdot f'(0) - \dots - f^{n-1}(0)$
Differentiation Laplace Domain	$(-t)^n \cdot f(t)$	$F^n(s)$
Integration Time Domain	$\int_0^t f(u) du$	$\frac{1}{s} \cdot F(s)$
Integration Frequency Domain	$\frac{1}{t} f(t)$	$\int_0^{\infty} F(u) du$
Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Initial Value Theorem $t \rightarrow 0$	$\lim_{t \rightarrow 0} f(t)$	$\lim_{s \rightarrow \infty} s \cdot F(s)$
Final Value Theorem $t \rightarrow \infty$	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} s \cdot F(s)$

Time Domain	Laplace Domain
$\delta(t)$	1
1 (Heavyside Step Function $\sigma(t)$)	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
t	$\frac{1}{s^2}$
$\sin(t)$	$\frac{1}{s^2+1}$
$t e^{at}$	$\frac{1}{(s-a)^2}$
$\frac{e^{at}-1}{a}$	$\frac{1}{s(s-a)}$
$\frac{e^{at}-at-1}{a^2}$	$\frac{1}{s^2(s-a)}$
$\frac{1}{2}t^2$	$\frac{1}{s^3}$
$\frac{1}{2}t^2 \cdot e^{at}$	$\frac{1}{(s-a)^3}$
$\frac{\sin(a \cdot t)}{a}$	$\frac{1}{s^2+a^2}$
$\cos(at)$	$\frac{s}{s^2+a^2}$
$\frac{\sinh(at)}{a}$	$\frac{1}{s^2-a^2}$
$t \cdot \cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
$\frac{1}{\sqrt{\pi t}}$	$\frac{1}{\sqrt{s}}$
$\frac{1-e^{at}}{t}$	$\ln\left(\frac{s-a}{a}\right)$
$\frac{\sin(at)}{t}$	$\arctan\left(\frac{a}{s}\right)$

Relationship between State Space Representation and System Function:

$$G(s) = \underline{c}^T (s\mathbf{I} - \underline{A})^{-1} \cdot \underline{B}$$

FIR Filter Design

Window Method

Window Type	Function	Peak Sidelobe Amplitude (Relative) in dB	Approximate Width of Main Lobe	Peak Approximation Error in dB
Rectangular	$w(n) = 1$	-13	$\frac{4\pi}{N+1}$	-21
Bartlett	$w(n) = 1 - \frac{2 n - \frac{N}{2} }{N}$	-25	$\frac{8\pi}{N}$	-25
Hann	$w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N}\right)$	-31	$\frac{8\pi}{N}$	-44
Hamming	$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$	-41	$\frac{8\pi}{N}$	-53
Blackman	$w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N}\right) + 0.08 \cos\left(\frac{4\pi n}{N}\right)$	-57	$\frac{12\pi}{N}$	-74

Window Type	Function
Kaiser	$w(n) = \frac{I_0 \left[\beta \sqrt{1 - \left(\frac{2n}{N-1} \right)^2} \right]}{I_0(\beta)}$ $I_0(x) = \sum_{k=0}^{\infty} \left(\frac{x^k}{2^k k!} \right)^2$ $A = -20 \log(\delta)$ $\Delta\omega = \omega_{\text{stop}} - \omega_{\text{pass}}$ $\beta = \begin{cases} 0.1102(A - 8.7) & A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 \leq A \leq 50 \\ 0.0 & A < 21 \end{cases}$ $N = \frac{A - 8}{2.285\Delta\omega}$

Optimal Method

Minimum order N_e of the filter for the optimal method:

$$N_e = \frac{2}{3} \log \left(\frac{1}{10\delta_{\text{pass}}\delta_{\text{stop}}} \right) \cdot \frac{\omega_{\text{stop}}}{\omega_{\text{stop}} - \omega_{\text{pass}}}$$

IIR Filter Design

Methods and their corresponding mapping rules

Replacing Derivatives by Differences:

$$s = \frac{z - 1}{T}$$

Bilinear Transform:

$$s = \frac{2}{T} \cdot \frac{z - 1}{z + 1} = \frac{2}{T} \cdot \frac{1 - z^{-1}}{1 + z^{-1}} \quad \Bigg| \quad \Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

Butterworth Filter

General equation

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad \Bigg| \quad |H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}$$

Identity

$$|H_c(s)|^2 = H_{c-1}(s) \cdot H_{c-1}(-s)$$

In case of 4 poles:

$$|H_c(s)|^2 = \frac{\Omega_c^2 \cdot \Omega_c^2}{(s - p_1)(s - p_2)(s - p_3)(s - p_4)}$$

Order of Butterworth Filter

$$N = \frac{1}{2} \frac{\log \left(\frac{1}{|H_c(j\Omega_{\text{pass}})|^2} - 1 \right) - \log \left(\frac{1}{|H_c(j\Omega_{\text{stop}})|^2} - 1 \right)}{\log \Omega_{\text{pass}} - \log \Omega_{\text{stop}}}$$

Convolution

Linear

$$y(t) = u(t) * g(t) = \int_{-\infty}^{+\infty} u(\tau) \cdot g(t - \tau) d\tau$$

$$y[k] = u[k] * g[k] = \sum_{\kappa=-\infty}^{+\infty} u[\kappa] \cdot g[k - \kappa]$$

Circular

$$y(t) = u(t) \circledast g(t) = \int_{t_0}^{t_0+T} u(\tau) \cdot g(t - \tau) d\tau$$

$$y[k] = u[k] \circledast g[k] = \sum_{\kappa=0}^{N-1} u[\kappa] \cdot g[k - \kappa]_N$$

Correlation

Autocorrelation

$$\phi_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n + m]$$

Cross-Correlation

$$\phi_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n + m]$$

DTFT

$$X_{\text{DTFT}}(j\omega) = \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega k T_a}$$

$$X_{\text{DTFT}}(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} x[k] e^{-jk\Omega}$$

$$\Omega = \omega T_a = \frac{\omega}{f_a} = \frac{2\pi f}{f_a}$$

$$x[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_{\text{DTFT}}(e^{j\Omega}) \cdot e^{jk\Omega} d\Omega$$

DFT

$$X_{\text{DFT}}[m] = \sum_{n=0}^{N-1} x[n] e^{-\frac{jn2\pi m}{N}} = \sum_{n=0}^{N-1} x[n] \cdot W_N^{nm}$$

$$W_N = e^{-\frac{j2\pi}{N}}$$

$$x[n] = \frac{1}{N} \sum_{m=0}^{N-1} X_{\text{DFT}}[m] \cdot W_N^{-nm}$$

Mathematical Basics

Addition Theorem:

$$\sin(\alpha) \cdot \cos(\beta) + \cos(\alpha) \cdot \sin(\beta) = \sin(\alpha + \beta)$$

Geometric Series:

$$\sum_{i=0}^{N-1} r^i = \frac{1 - r^N}{1 - r}$$

Integration by Parts:

$$\int_a^b f'(x) \cdot g(x) dx = \left[f(x) \cdot g(x) \right]_a^b - \int_a^b f(x) \cdot g'(x) dx$$